How a Sustainable Maritime Traffic Can Shape the California's Kelp Forests

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January 21, 2026

Abstract

This article studies how maritime regulation and vessel traffic affect coastal ecosystems by focusing on kelp forests along the California–Baja California coastline. We combine quarterly satellite measures of kelp canopy and biomass at 30m resolution with vessel trajectories reconstructed from AIS signals (U.S. Coast Guard). Our first design exploits California's Ocean-Going Vessel Fuel Rule, implemented in July 2009, which induced changes in nearshore routing. Using a difference-in-differences framework comparing kelp located inside the regulated zone to nearby kelp forests in Baja California, we find a significant post-2009 increase in kelp outcomes in treated areas. We then use the 2011 boundary modification around the Channel Islands, which re-concentrated traffic in specific corridors, to relate shipping exposure to kelp outcomes. In areas where vessel passage intensity rises after 2011, kelp canopy and biomass decline, consistent with ecological costs of maritime traffic near sensitive habitats. JEL: Q57, R41, F18

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[†]We would like to thank David Nagy, Sandra Poncet, Pamina Koenig, Antoine Bouet, Aicha Hasni, Elise Labauvie, Isabelle Chort and Christophe Maes for valuable comments and discussions. We also thank all the participants of several seminars (RITM, BSE, CEEM) particularly Raphael Chiappini, Tanguy Bernard and Sebastien Desbureaux and as well as participants of 29th Annual Conference of the EAERE. We are grateful for the excellent working conditions provided by the Atlantic Port of Bordeaux, Bordeaux School of Economics and to Pompeu Fabra University. Finally, we thank the Region Nouvelle-Aquitaine (project PORTeur) for providing all the funds necessary for this research.

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1 Introduction

Ocean-going vessels generate substantial externalities. The consequences of air pollution related to shipping for human health are now well documented (e.g. Klotz and Berazneva, 2022; Hansen-Lewis and Marcus, 2022), but evidence on the impacts of maritime activity on marine ecosystems remains comparatively limited at the spatial scales relevant for policy. This paper investigates this question by quantifying how a major coastal shipping regulation and the associated reconfiguration of traffic patterns relate to the dynamics of kelp forests along the California coast.

Kelp forests are large biogenic habitats that support coastal biodiversity and provide multiple ecosystem services. As a foundation species (Paine, 1969; Hale and Koprowski, 2018), kelp creates habitat structure that functions as nursery grounds and supports complex food webs (Christie et al., 2009). Recent global accounting exercises suggest that kelp ecosystems contribute sizable benefits through fisheries production, nutrient cycling, and carbon sequestration (Eger et al., 2023). Because kelp can respond quickly to changing local conditions through rapid growth and recruitment (Krumhansl et al., 2016; Carr and Reed, 2019), it offers an informative setting to study ecosystem responses to changes in nearshore stressors over a relatively short horizon.

We study California's Ocean-Going Vessel Fuel Rule, which established a regulated zone in 2009 and later modified the boundary in southern California in 2011. The policy was designed to reduce sulfur-related emissions by requiring the use of low-sulfur marine distillate fuels within the zone. Importantly for our setting, the rule altered vessel routing decisions, thereby changing the spatial footprint of maritime activity near kelp forests, an effect documented by Klotz and Berazneva (2022). These routing adjustments provide quasi-experimental variation that can be exploited to learn about the sensitivity of kelp forests to maritime exposure.

Our contribution is twofold. First, we combine quarterly satellite-based measures of kelp canopy and biomass with a difference-in-differences design that compares kelp located within the regulated zone to kelp forests in Baja California that share oceanographic continuity but are not subject to the California rule. This approach isolates changes in kelp outcomes that occur differentially inside the regulated area after 2009, net of time-invariant local characteristics and common temporal shocks. We find a robust post-2009 increase in kelp canopy and biomass across several treated locations (Los Angeles–Long Beach, San Diego, and Hueneme). We interpret these estimates as evidence that changes in nearshore maritime pressure induced by the regulation are associated with measurable improvements in kelp outcomes.

Second, we link kelp outcomes to direct measures of shipping activity derived

from AIS vessel tracks. We build exposure measures that capture both the frequency of vessel passages and the intensity of traffic (distance traveled) near kelp forests. We then exploit the 2011 boundary modification around the Channel Islands—a change that re-concentrated traffic in specific corridors (Klotz and Berazneva, 2022)—to estimate how the kelp–traffic relationship shifts when vessels return to, or intensify in, particular lanes. In areas where traffic increases after 2011, kelp canopy and biomass decline, indicating a tangible ecological cost of concentrated maritime traffic in proximity to kelp habitats.

This analysis relates to three strands of work. First, it complements research on international economic activity and environmental outcomes by bringing marine "foundation species" into the empirical environmental-economics toolkit (e.g., Berman et al., 2023; Abman and Lundberg, 2020; Carreira et al., 2024; Nedoncelle et al., 2024). Second, it complements experimental and site-specific ecological studies on shipping-related stressors (e.g., Johansson et al., 2012) by providing a spatially comprehensive, policy-relevant analysis that follows a regulatory change. Third, it connects to the literature evaluating emission control policies in maritime settings by documenting ecosystem responses in coastal waters, beyond human-health endpoints (e.g., Klotz and Berazneva, 2022; Hansen-Lewis and Marcus, 2022).

2 Background

The California–Baja California coastline hosts extensive kelp forests dominated by canopy-forming species (giant kelp, Macrocystis pyrifera, and bull kelp, Nereocystis luetkeana). These species form surface (or near-surface) canopies that are detectable with satellite imagery (Bell et al., 2023).

Kelp performance is shaped by ocean temperature, nutrient availability, water motion, and light, as well as by localized stressors that can damage fronds, inhibit photosynthesis, or alter community structure (Carr and Reed, 2019). Climatic variability is then a first-order driver of kelp outcomes. Ocean climate oscillations influence sea surface temperatures, salinity, and nutrient availability over multi-year cycles (Castorani et al., 2022). Warm-water events can sharply reduce canopy extent, and the decline observed since 2014 has been linked to prolonged warm conditions and cascading ecological interactions (Rogers-Bennett and Catton, 2019; see also Krumhansl et al., 2016). These mechanisms motivate the inclusion of sea surface temperature and wave intensity controls in our empirical models.

Maritime activity can affect kelp through several channels. Vessel operations generate discharges (greywater and blackwater) and associated chemical and nutrient loads that can contribute to eutrophication and oxygen depletion (reviewed

in Lindgren et al., 2016), with potential consequences for coastal vegetation and algal community composition (e.g., Filbee-Dexter and Wernberg, 2018). Shipping can also introduce contaminants through antifouling practices. Biocidal compounds historically used to reduce hull fouling can enter coastal waters and affect marine organisms (Dafforn et al., 2011; Cima and Varello, 2022), with laboratory evidence that antifouling compounds can inhibit kelp photosynthesis and growth (Johansson et al., 2012). Finally, maritime traffic can influence nearshore physical conditions (wake, turbulence, localized disturbance) and can concentrate stressors along heavily trafficked corridors.

A distinct, potentially important channel is the introduction of non-native species via hull fouling and ballast water. The literature has documented introductions linked to shipping, with consequences that vary across species and contexts (Williams and Smith, 2007). Evidence on impacts of particular invaders on native kelp is mixed, including work on Sargassum muticum (Ambrose and Nelson, 1982; Britton-Simmons, 2004; Smith, 2015) and more recent assessments of Undaria pinnatifida (McHaskell, 2024). Ballast water can also transport grazers and pathogens with ecosystem-wide effects (e.g., Phinney et al., 2001).

While our empirical designs do not identify the relative importance of these channels, they motivate the hypothesis that changes in the spatial distribution and intensity of shipping activity can matter for kelp outcomes. California's Ocean-Going Vessel Fuel Rule introduced a regulated zone extending 24 nautical miles off the coast starting in 2009 that enables to test this hypothesis. Covered vessels must use low-sulfur marine distillate fuels within the zone. Compliance increases operating costs within the regulated area and can induce changes in routing and the spatial distribution of traffic. Independent evidence shows that vessels adjusted routes following implementation, reducing nearshore passages in some areas and changing approach paths to ports (Klotz and Berazneva, 2022). Finally, in 2011, the southern boundary was modified around the Channel Islands, affecting where ships concentrate as they approach ports and creating additional variation in maritime exposure (Klotz and Berazneva, 2022).

3 Empirical strategy

This section presents the main difference-in-differences design used to estimate changes in kelp canopy and biomass associated with the 2009 implementation of the regulated zone. We use quarterly data from 2007 to 2011, covering multiple treated kelp regions along the California coast and a control region in Baja California.

We estimate a two-way fixed effects difference-in-differences model:

$$Kelp_{it} = \alpha + \beta\, California_i\, Post2009_t + Z_{it} + f_i + f_t + \varepsilon_{it}, \tag{1}$$

where $Kelp_{it}$ denotes kelp outcomes measured at the cell level i in quarter t (kelp canopy in m² and kelp biomass in kg). $California_i$ equals one for kelp cells located within the 2009 regulated zone and zero otherwise. $Post2009_t$ equals one from 2009Q3 onward and zero otherwise. Z_{it} includes time-varying environmental and economic controls described below. Cell fixed effects f_i absorb time-invariant local characteristics; quarter fixed effects f_t capture common shocks.

Because the dependent variables include many zeros (cells with no kelp in a given quarter), we estimate Eq. (1) using Poisson pseudo-maximum likelihood (PPML), which is well-suited to non-negative outcomes with heteroskedasticity and zeros (e.g., Santos Silva and Tenreyro, 2006). Standard errors are clustered at the kelp-cell level.

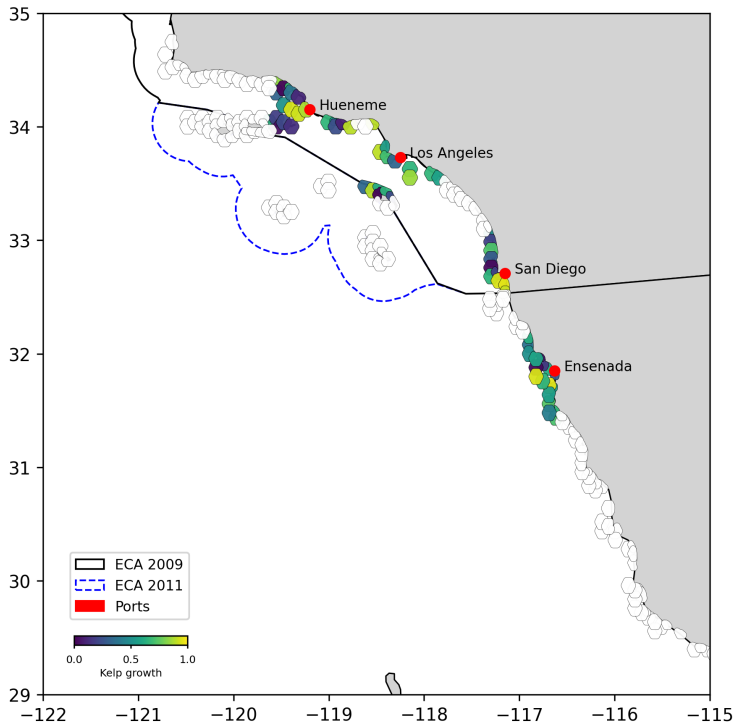
To assess dynamics and pre-trends, we estimate an event-study specification:

$$Kelp_{it} = \alpha + \sum_{\ell} \beta_\ell Area_{i,t-\ell} + Z_{it} + f_i + f_t + \varepsilon_{it}, \tag{2}$$

where ℓ indexes quarters relative to implementation ($\ell = 0$ at 2009Q3). Coefficients for $\ell < 0$ provide a falsification test for differential pre-trends; coefficients for $\ell > 0$ trace the evolution of post-policy effects.

3.1 Sample

Map 1 presents the different group of kelp studied in the treated groups and in the control group.



Note: The different hexagons represent the aggregated kelp canopy. The black line represents the limit of the ECA enforced in 2009. The blue dashed line represents the 2011 extension (studied in the last section). Projections are under EPSG:4326 - WGS 84 coordinate reference system.

Figure 1: Kelp in the ECA and in Baja California

We focus on kelp forests near three port areas inside the regulated zone: Los Angeles–Long Beach, San Diego, and Hueneme. The Los Angeles–Long Beach complex is our baseline treated location. It is the main gateway for trans-Pacific trade on the U.S. West Coast and, in 2010, accounted for 28% of the total value of U.S. maritime trade (US, 2010). Its scale, the density of port approaches, and the concentration of deep-draft commercial traffic make it the setting where policy-induced changes in routing and nearshore vessel activity are most likely to be quantitatively important. Consistent with this, Klotz and Berazneva (2022) document that vessels adjusted their routes following the Ocean-Going Vessel Fuel Rule, a behavioral response that is particularly relevant in a port complex of this size. For each treated port area, we include all kelp pixels within a 50 km coastal buffer. This radius is chosen to capture kelp forests plausibly exposed to changes in shipping lanes and port approaches, while reducing sensitivity to highly local

dynamics related to sedimentation, recruitment, and small-scale coastal heterogeneity.

San Diego and Hueneme provide complementary treated settings that strengthen external validity and serve as robustness checks. Both ports host substantial commercial traffic and lie on major coastal routes, but they differ in size, function, and coastal configuration. San Diego is the southernmost treated port and is closest to Baja California, which is useful for two reasons. First, the shorter geographic distance to the control region helps alleviate concerns that results are driven by abrupt differences in oceanographic regimes rather than by changes in maritime exposure. Second, its position near the international border provides a natural "border-proximate" treated area. Hueneme, while smaller than Los Angeles–Long Beach, sits near the approaches to the Santa Barbara Channel and the Channel Islands Passage. This location matters because the 2011 boundary modification was explicitly designed to influence routing and to move traffic outside in this corridor (Klotz and Berazneva, 2022), which makes Hueneme and nearby kelp forests informative for assessing whether changes in vessel presence translate into measurable ecological responses.

The choice of these three treated port areas also reflects an identification strategy that does not rely on a single local comparison. If the estimated effect around Los Angeles–Long Beach were driven by idiosyncratic local factors—for example, a localized ecological shock, a change in coastal management unrelated to shipping, or unobserved coastal development occurring contemporaneously with the policy—it would be less likely to replicate in two additional port areas with different geography and different baseline traffic intensity. By estimating the same specification across Los Angeles–Long Beach, San Diego, and Hueneme, we can assess whether the post-2009 differences are systematically associated with treatment status rather than with a single-site coincidence. This multi-site design also aligns with the broader objective of the paper: to provide evidence that maritime regulation can shape nearshore ecosystems in economically relevant coastal settings, rather than documenting a single case-study effect.

As a control group, we use kelp near the port of Ensenada in Baja California. Ensenada is a deep-water commercial port and the principal maritime gateway for Baja California; it receives international vessel traffic and is embedded in the same Pacific shipping network as Southern California, but it lies outside California's fuel regulation (and there is no equivalent policy in Mexico). This combination is central for our difference-in-differences design: Ensenada is not a remote, low-activity coastline, but rather a port area that is plausibly exposed to similar macro-level fluctuations in maritime trade and shipping demand, while remaining untreated by the California policy. These considerations motivate Ensenada as a natural benchmark for the counterfactual evolution of kelp forests in the absence

of the policy.

Finally, we use complementary checks to address the concern that treated and control areas may differ in ways that could generate divergent trends even absent the policy. First, the event-study specification in Eq. (2) provides a direct test of differential pre-trends; coefficients prior to 2009 indicate that treated kelp areas were not evolving differently from Ensenada before implementation. Second, we propose a robustness exercise that does not rely on Ensenada alone, by using a Synthetic DID. By definition, this strategy reproduces the pre-period path closely.

3.2 Controls

Kelp is affected by multiple interacting environmental drivers, including temperature and water motion (Carr and Reed, 2019; Castorani et al., 2022). We include sea surface temperature (SST) and wave intensity as key physical controls, and we proxy coastal human activity using nighttime lights to capture time-varying local pressures potentially correlated with kelp outcomes (e.g., eutrophication risks discussed by Filbee-Dexter and Wernberg, 2018).

Sea surface temperature is sourced from MODIS monthly composites and aggregated to quarters. SST is a central determinant of kelp dynamics and is also informative about broader ocean conditions (e.g., Yang et al., 2023; Dunstan et al., 2018). Wave intensity is drawn from ERA5 (Hersbach et al., 2023) and measured as the average height of the highest third of waves generated by wind and swell, aggregated to quarters; it captures physical stress associated with storms and strong sea states. Economic activity is proxied using nighttime lights data (following Li et al., 2020).

3.3 Data

Kelp outcomes come from Bell et al. (2023), which provides quarterly cloud-free Landsat-based measures of floating kelp canopy at 30m resolution along the Pacific coast. We use two variables: (i) kelp canopy area within each pixel (m^2) and (ii) kelp canopy biomass (kg) inferred from validated relationships between surface reflectance and in situ canopy and biomass measurements (Bell et al., 2023). Satellite detection captures surface (or near-surface) canopy and does not observe subsurface kelp.

4 Results

4.1 Restoration of the Kelp forest

Table (1) reports estimates of Eq. (1) when kelp canopy is the dependent variable. Across treated regions, the post-2009 interaction term is positive and statistically significant. For the Los Angeles–Long Beach area, the PPML coefficient of 0.351 implies an increase in expected kelp canopy of approximately $100 \cdot (\exp(0.351) - 1) \approx 42\%$ relative to the control group after implementation. For San Diego and Hueneme, the estimated increases are approximately 54% and 26%, respectively.

	LA	San Diego	Hueneme
Diff-in-Diff Effect	0.351***	0.43***	0.231***
	(0.0871)	(0.0479)	(0.06)
Temperature	-0.0258	-0.474***	0.275***
	(0.0507)	(0.0378)	(0.0906)
Sea weight	-0.4093***	-0.2615***	-0.6828***
	(0.0149)	(0.0221)	(0.0823)
Econ	0.0225	-0.464***	-0.404***
	(0.0259)	(0.0220)	(0.0662)
Constant	10.03***	21.55***	11.63***
	(0.815)	(0.663)	(2.437)
Time FE	✓	✓	✓
Kelp FE	✓	✓	✓
Observations	105423	180743	44625
Log-Likelihood	-5372934.8	-8840835.4	-3129865.0
Pseudo R2	0.518	0.503	0.516

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Eq. (1) estimated by PPML with kelp canopy (area) as the dependent variable at the pixel level. Treated groups are Los Angeles, San Diego, and Hueneme; the control group is Ensenada. The sample covers 2007–2011 (quarterly). All covariates are in logs. Standard errors are clustered at the kelp-cell level.

Table 1: Kelp canopy growth after 2009

Table (2) repeats the analysis with kelp biomass as the dependent variable. The post-2009 interaction remains positive and significant. In the Los Angeles–Long

Beach area, the coefficient of 0.381 implies an increase in expected biomass of about $100 \cdot (\exp(0.381) - 1) \approx 46\%$ relative to the control group after implementation.

	LA	San Diego	Hueneme
Diff-in-Diff Effect	0.381***	0.45***	0.22***
	(0.0858)	(0.0476)	(0.06)
Sea surface temperature	-0.0298	-0.502***	0.295***
	(0.0501)	(0.0374)	(0.0906)
Sea surface weight	-0.4150***	-0.2631***	-0.6653***
	(0.0148)	(0.0219)	(0.0826)
Economic Activity	0.0302	-0.480***	-0.410***
	(0.0259)	(0.0213)	(0.0663)
Constant	12.07***	24.24***	13.08***
	(0.808)	(0.652)	(2.439)
Time FE	✓	✓	✓
Kelp FE	✓	✓	✓
Observations	105423	180743	44625
Log-Likelihood	-39124235.0	-63920371.9	-22690642.8
Pseudo R2	0.519	0.503	0.522

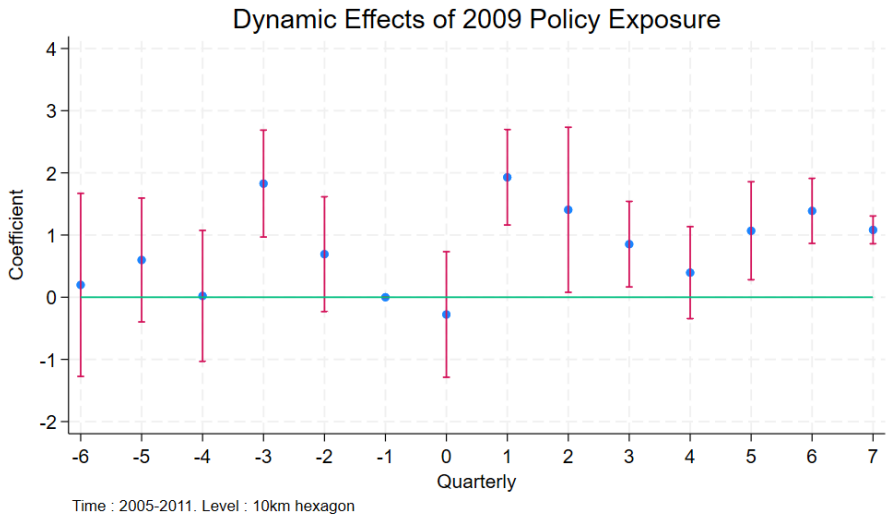
Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
Eq. (1) estimated by PPML with kelp biomass as the dependent variable at the pixel level. Treated groups are Los Angeles, San Diego, and Hueneme; the control group is Ensenada. The sample covers 2007–2011 (quarterly). All covariates are in logs. Standard errors are clustered at the kelp-cell level.

Table 2: Kelp biomass growth after 2009

These estimates are consistent with the interpretation that the policy-induced reduction or redistribution of nearshore maritime pressure is associated with improved kelp outcomes in the regulated zone. The environmental controls enter with signs consistent with kelp biology: higher temperatures reduce kelp outcomes and higher wave intensity is associated with lower canopy and biomass, consistent with physical damage mechanisms documented in the kelp literature (e.g., Carr and Reed, 2019).

4.2 Dynamic effects

Figure (2) reports event-study estimates from Eq. (2) for the Los Angeles–Long Beach region. The pre-policy coefficients are close to zero and generally statistically indistinguishable from zero, supporting the identifying assumption of parallel pre-trends. Post-policy coefficients are positive through the end of the sample, indicating that the estimated post-2009 increase reflects sustained differences rather than a short-lived deviation. Seasonal kelp dynamics can create quarter-to-quarter variation in measured canopy and biomass (e.g., Cavanaugh et al., 2019), which motivates interpreting the event-study in terms of the broader post-policy pattern rather than any single quarter.



Note: This graph shows quarterly dynamic effects on kelp biomass (log scale) in the treated group (Los Angeles) compared to the control group (Ensenada), using the DID method. All covariates are included in the model.

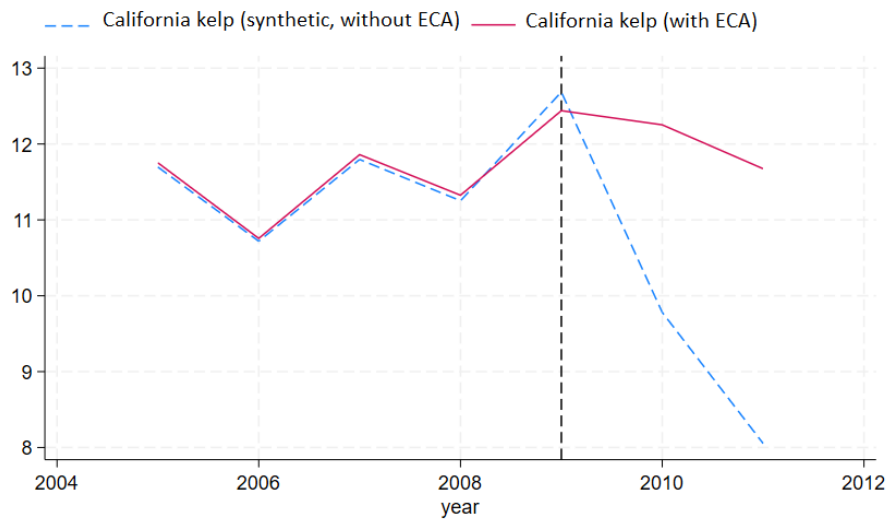
Figure 2: Dynamic effect for kelp in L.A.

4.3 Synthetic difference-in-differences

To complement the port-based difference-in-differences results, we implement a synthetic control exercise that (i) uses a different spatial unit and (ii) expands the counterfactual beyond a single control port. We aggregate kelp biomass from 30m $\times$ 30m pixels into 10 km hexagonal cells and construct an annual series for each

hexagon. This aggregation reduces high-frequency noise driven by local patch dynamics and seasonal measurement variation, and it aligns naturally with our objective of testing whether the post-2009 divergence is visible along the broader coastline rather than only in the immediate vicinity of major ports.

The treated sample for this exercise consists of coastal hexagons located inside the 2009 regulated boundary but outside the port buffers already analyzed in the main DID specifications; we also exclude hexagons that fall within the 2011 boundary modification to keep the exercise focused on the 2009 policy regime. The donor pool is composed of coastal hexagons on the Mexican shoreline (Figure 7). We then construct a synthetic counterpart for the treated coastline as a weighted combination of donor hexagons chosen to reproduce the treated pre-2009 trajectory as closely as possible. This design provides an out-of-sample check that the baseline findings are not driven by the particular choice of Ensenada as the comparison location.



Note: This figure displays the kelp biomass trends (summed by hexagon and year) for the treated hexagons located within the 2009 ECA zone—excluding those already analyzed—compared to a Synthetic California kelp biomass (constructed without ECA) based on untreated hexagons on the Mexican coast. These spatial units correspond to the transparent hexagons shown in Figure 1.

Figure 3: Synthetic Control for unstudied kelp area

Figure 3 reports the resulting treated and synthetic series. The pre-2009 fit is close, indicating that the weighted donor combination reproduces the level and

evolution of biomass in the treated coastline before implementation. After 2009, the series diverge: biomass remains stable (or declines only moderately) in treated California hexagons, while the synthetic counterpart declines sharply. In the synthetic control framework, this post-2009 divergence is precisely the object of interest, it shows that in a counterfactual world without ECA, kelp would be in less favorable conditions. We interpret this exercise as supportive evidence that the post-2009 improvement documented in the main DID results is not confined to the port areas and is robust to an alternative construction of the counterfactual coastline.

5 The Narrow Corridor where Vessels Come Back

This section uses AIS vessel tracks to construct direct measures of maritime exposure and relates these measures to kelp outcomes in the period after implementation and before the 2014–2016 marine heatwave that is widely viewed as a major driver of kelp declines (Krumhansl et al., 2016; Rogers-Bennett and Catton, 2019). The objective is to move from a reduced-form post-policy comparison to a specification that explicitly links kelp dynamics to observed changes in nearby vessel traffic.

5.1 Exposure of kelp to vessels

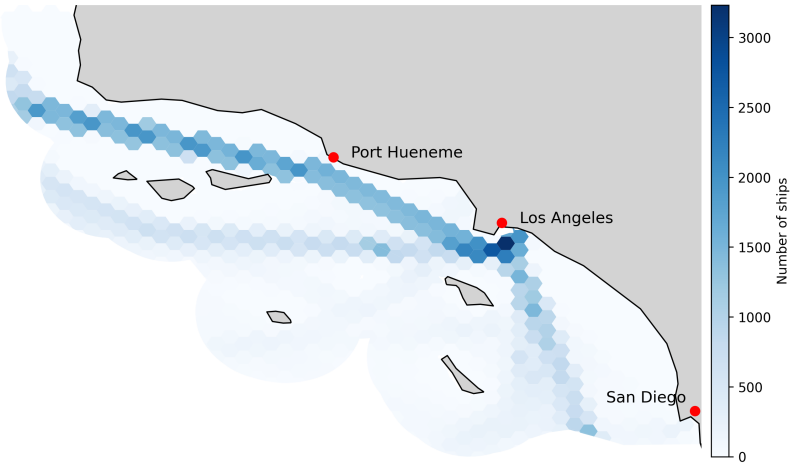
To reconstruct vessel trajectories, we overlay a uniform grid of 10-kilometer hexagonal cells covering the original 2009 regulated zone and the 2011 extension. Trajectories are derived from georeferenced ship positions recorded by the Automated Identification System (AIS), made available by the U.S. Coast Guard.[1] We focus on four major cargo vessel classes—tankers, container ships, general cargo ships, and bulk carriers.

For each hexagon and quarter, we compute two measures of shipping intensity: (i) the number of unique vessels traversing the cell ("count"), and (ii) the total distance traveled within it ("length"). We assign these indicators to kelp aggregation hexagons along the coast by computing exposure from the nearest traffic cells and weighting by inverse distance. This yields a continuous measure that increases with both proximity to shipping lanes and traffic intensity.

[1] AIS data accessed via MarineCadastre.gov, jointly maintained by the U.S. BOEM and NOAA.

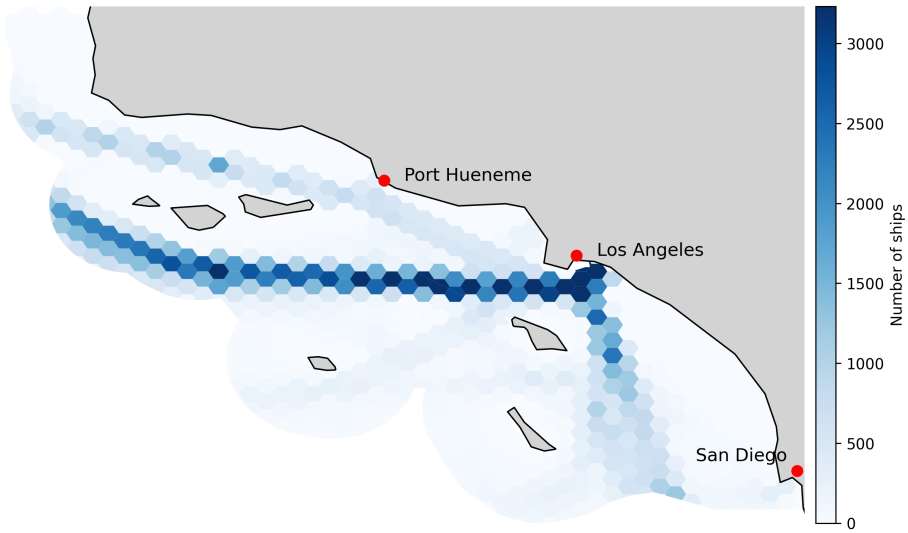
5.2 Extension of the ECA in 2011

The 2011 boundary modification around the Channel Islands altered where vessels concentrate as they approach Southern California ports and contributed to a reallocation of traffic toward specific corridors (Klotz and Berazneva, 2022). Figures (4) and (5) illustrate the spatial distribution of traffic exposure around southern California in the two periods.



Note: We measure the number of ships passing through each hexagonal cell in Southern California in 2009 to characterize shipping patterns at the start of the regulated period.

Figure 4: Exposure of Kelp to Maritime Traffic in 2009



Note: This map shows the number of ships crossing each hexagonal cell in 2011, immediately prior to the boundary modification in southern California.

Figure 5: Exposure of Kelp to Maritime Traffic in 2011

We provide a complementary time-series depiction of the redistribution of traffic between the 2009 zone and the extension area in Appendix C.

5.3 Empirical strategy

To assess whether the relationship between traffic exposure and kelp outcomes changed after the 2011 boundary modification, we estimate:

$$Kelp_{it} = \alpha + \beta Exposure_{it} + \gamma Policy_t Exposure_{it} + Z_{it} + f_i + f_t + \varepsilon_{it}, \tag{3}$$

where $Exposure_{it}$ is the continuous measure of vessel traffic near kelp (constructed from AIS tracks), $Policy_t$ equals one after 2011 and zero otherwise, and f_i and f_t are kelp-cell and time fixed effects. With $Exposure_{it}$ in logs and PPML estimation, coefficients can be interpreted as elasticities of expected kelp outcomes with respect to exposure (see Santos Silva and Tenreyro, 2006 for properties of PPML under log-linear mean specifications).

5.4 Results

Table (3) reports estimates using logged length exposure. The interaction term is negative and statistically significant for both biomass and canopy. Interpret-

ing the coefficients as elasticities, the post-2011 elasticity of kelp canopy with respect to exposure is $\beta + \gamma = -0.380 - 0.747 \approx -1.13$, implying that a 1% increase in shipping exposure is associated with about a 1.13% decrease in expected kelp canopy after 2011 in the affected areas. The implied elasticity for biomass is similar ($-0.359 - 0.744 \approx -1.10$). Appendix D shows that results remain consistent when exposure is measured using vessel counts rather than distance traveled.

	Biomass	Canopy
Length exposure (ln)	-0.359	-0.380
	(0.520)	(0.530)
Policy * Length exposure (ln)	-0.744**	-0.747**
	(0.338)	(0.045)
Sea surface temperature (ln)	-0.490	-0.418
	(1.516)	(1.552)
Sea weight intensity (ln)	-0.266**	-0.270**
	(0.1132)	(0.1149)
Economic activity (ln)	0.105	0.106
	(0.145)	(0.147)
Time FE	✓	✓
kelp FE	✓	✓
Observations	702	702
Log-Likelihood	-74977217.3	-10229451.1
Pseudo R2	0.831	0.829

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
The dependent variable is the sum of kelp biomass (in kg) and area (in m^2) within each hexagonal cell located in the 2009 zone and the 2011 extension zones, measured quarterly. Exposure is defined as the total distance traveled by ships within each hexagonal cell in proximity to kelp forests. All covariates are in logs. The model includes time fixed effects and kelp-cell fixed effects and is estimated by PPML.

Table 3: 2011 expansion impacts on zone affected in 2009 and 2011

6 Conclusion

If in any country a forest was destroyed, I do not believe as many species of animals would perish as would here from the destruction of

kelp. Darwin, 1839

The benefit of kelp for human date from pre-historic time; Erlandson et al. (2007) even propose a 'Kelp Highway Hypothesis' explaining that human migration 20'000 years ago depended on this resource. However, the "highway of kelp" is now taken by giant vessels that leave in their wake maritime pollution. We have shown in this article that when traffic is relatively decreasing, the growth of the kelp canopy increases in the California ECA.

This indicates that the Ocean-Going Vessel Fuel Rule has succeeded not only in protecting human health, as shown by Klotz and Berazneva (2022), but also in protecting the underwater environment in places where traffic has decreased. Our analysis is obviously limited to a particular area and period, but it has implications for both economic analysis and maritime policy.

Firstly, since the international trade literature now examines the "hidden costs" of globalization, we propose here another topic of interest, as kelp forests play an important role in various areas, from fishing to alginate, not to mention carbon sequestration in oceans.

Second, this study supports the objective of the International Maritime Organization (IMO), which has been pivotal in enforcing essential regulations, including emission control areas over the past decade. We show that these areas may have an effect beyond their impact on air pollution, and then could have a role to play in the future in promoting sustainable international trade that respects the underwater canopy.

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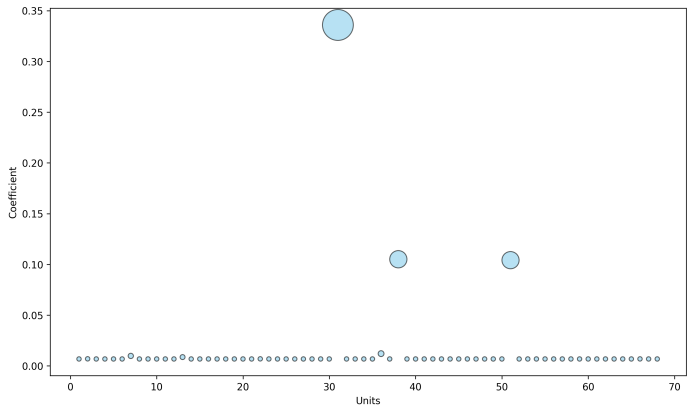
Appendix A: Descriptive Statistics

Appendix B: Synthetic Differences-in-difference

This appendix reports a synthetic control exercise designed to complement the baseline difference-in-differences strategy and to reduce dependence on a single control location. The analysis is conducted on a 10 km hexagonal grid. We aggregate kelp biomass from 30m $\times$ 30m pixels to hexagons and construct an annual series for each hexagon.

Treated units are coastal hexagons located within the 2009 regulated boundary. To ensure that this exercise provides evidence distinct from the main port-based estimates, we exclude hexagons surrounding the ports of Los Angeles–Long Beach, San Diego, and Hueneme, which are already analyzed in the main text. We also omit hexagons located within the area corresponding to the 2011 boundary modification. The donor pool consists of coastal hexagons along the Mexican coastline. Figure 7 displays the treated and donor units used in the synthetic control construction.

The synthetic control is constructed as a weighted combination of donor hexagons selected to replicate the treated pre-2009 biomass trajectory as closely as possible. The resulting weights are reported in Figure 6.



Note: The dependent variable is the sum of biomass in each hexagon's cells.

Figure 6: Synthetic control (weight)

A small number of donor hexagons receive most of the weight, which indicates that these locations are the most informative for matching the treated pre-period path. This feature is common in synthetic control applications.

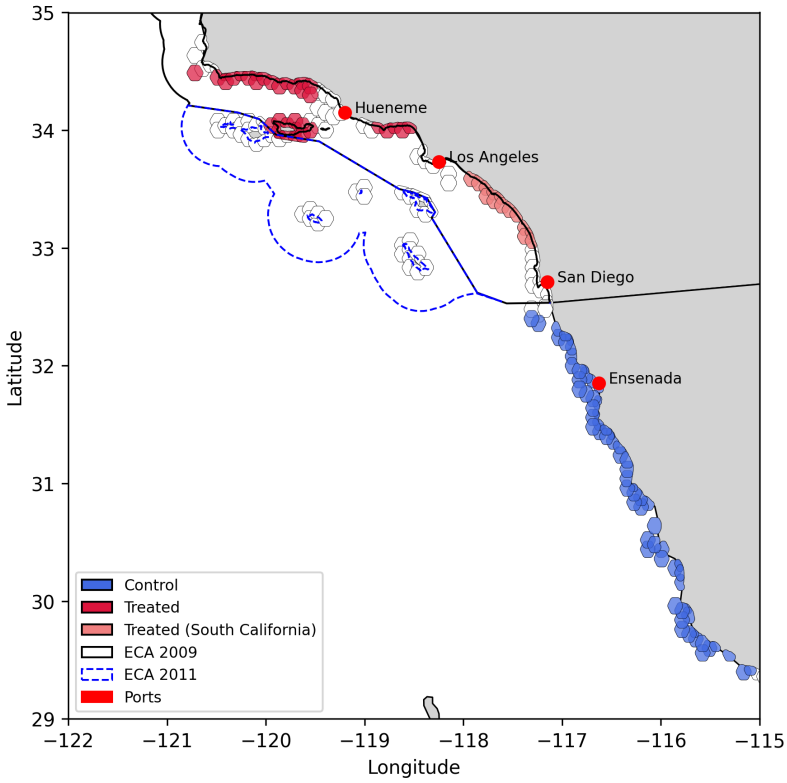


Figure 7: Map for the Synthetic Control Analysis

Table 4 reports the estimated post-2009 effect for two treated samples: (1) all treated hexagons shown in red in Figure 7, and (2) a treated sample excluding the southern California subset (light red in Figure 7).

	(1)	(2)
	Biomass (ln)	
Synthetic Control	1.391**	1.911**
	(0.642)	(0.753)
N	819	721

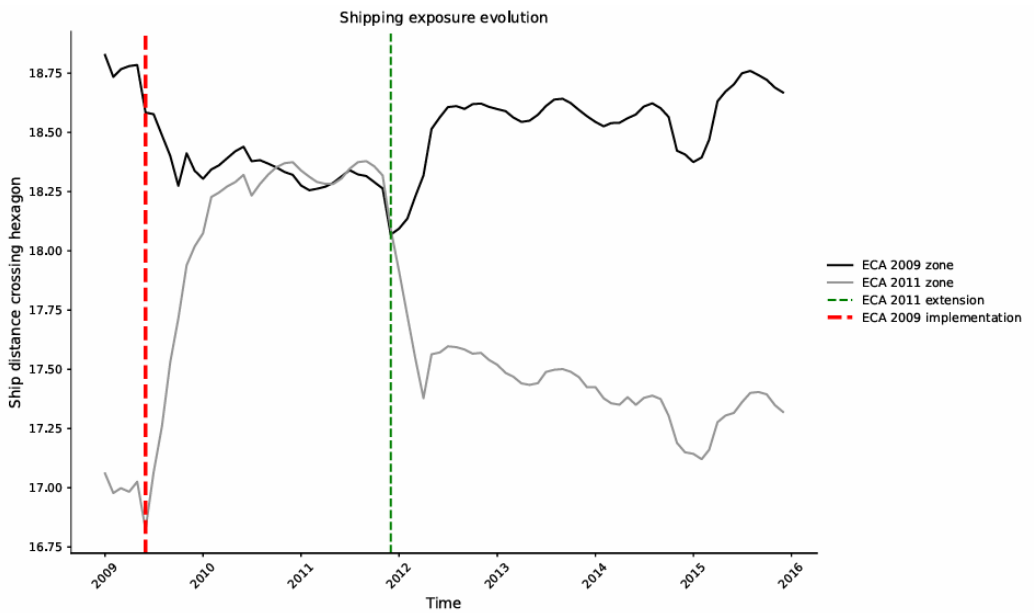
Table 4: SDID Robustness checks

In both cases, the estimated effect is positive and statistically significant, indicating that biomass in treated California hexagons evolves more favorably after

2009 than in their synthetic counterparts constructed from the Mexican donor pool. Figure 3 in the main text shows the same result in levels: the synthetic series closely tracks the treated series prior to 2009 and then falls markedly afterwards, while the treated series remains comparatively stable. The post-2009 decline in the synthetic series reflects the realized evolution of the donor locations that best reproduce the treated pre-period trajectory.

Appendix C: After 2011 extension

Figure 8 measures the total distance traveled by vessels within each hexagonal cell, comparing the 2009 zone and the area corresponding to the 2011 extension. The series illustrates a redistribution of traffic across these areas over time, consistent with route adjustments and the boundary modification documented in Klotz and Berazneva (2022).



Note: This graph shows the distance traveled by ships in each hexagonal cell near kelp forests, distinguishing between cells located in the 2009 zone and those in the 2011 extension. The data is presented monthly from 2009 to 2016.

Figure 8: Evolution of shipping patterns

Appendix D: robustness exposure measure

We replace the exposure measure based on total distance traveled with the number of ships crossing each hexagonal cell. The results remain consistent with the main findings.

	Biomass	Area
Policy * Count exposure (ln)	-0.0894**	-0.0908**
	(0.0371)	(0.0379)
Count exposure (ln)	-0.0623	-0.0655
	(0.0761)	(0.0770)
Sea surface temperature (ln)	-0.442	-0.365
	(1.535)	(1.572)
Sea weight intensity (ln)	-2.766**	-2.811**
	(1.130)	(1.146)
Economic activity (ln)	0.125	0.127
	(0.144)	(0.146)
Constant	15.23***	13.02***
	(4.547)	(4.647)
Time FE	✓	✓
kelp FE	✓	✓
Observations	702	702
Log-Likelihood	-74892022.1	-10211572.1
Pseudo R2	0.831	0.829

Notes: Standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.
The dependent variable is the sum of kelp biomass (in kg) and area (in m^2) within each hexagonal cell located in the 2009 zone and the 2011 extension zones, measured quarterly. Exposure is defined as the total number of ships within each hexagonal cell in proximity to kelp forests. All covariates are in logs. The model includes time fixed effects and kelp-cell fixed effects and is estimated by PPML.

Table 5: Shipping exposure (count, sum) impact on kelp: 2009 and 2011 zone